

```
In [2]: !pip install numpy
```

```
Requirement already satisfied: numpy in c:\users\mohan\anaconda3\lib\site-packages (1.24.3)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
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WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
```

```
In [3]: !pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\mohan\anaconda3\lib\site-packages (0.11.2)
Requirement already satisfied: scipy>=1.0 in c:\users\mohan\anaconda3\lib\site-packages (from seaborn) (1.11.2)
Requirement already satisfied: pandas>=0.23 in c:\users\mohan\anaconda3\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: numpy>=1.15 in c:\users\mohan\anaconda3\lib\site-packages (from seaborn) (1.24.3)
Requirement already satisfied: matplotlib>=2.2 in c:\users\mohan\anaconda3\lib\site-packages (from seaborn) (3.4.3)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib>=2.2->seaborn) (1.3.1)
Requirement already satisfied: pyparsing>=2.2.1 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib>=2.2->seaborn) (3.0.4)
Requirement already satisfied: pillow>=6.2.0 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib>=2.2->seaborn) (9.5.0)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib>=2.2->seaborn) (2.8.2)
Requirement already satisfied: cycler>=0.10 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib>=2.2->seaborn) (0.10.0)
Requirement already satisfied: six in c:\users\mohan\anaconda3\lib\site-packages (from cycler>=0.10->matplotlib>=2.2->seaborn) (1.16.0)
Requirement already satisfied: pytz>=2020.1 in c:\users\mohan\anaconda3\lib\site-packages (from pandas>=0.23->seaborn) (2021.3)
Requirement already satisfied: tzdata>=2022.7 in c:\users\mohan\anaconda3\lib\site-packages (from pandas>=0.23->seaborn) (2024.1)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
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WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
```

```
In [4]: !pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\mohan\anaconda3\lib\site-packages (2.2.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\mohan\anaconda3\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\mohan\anaconda3\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: pytz>=2020.1 in c:\users\mohan\anaconda3\lib\site-packages (from pandas) (2021.3)
Requirement already satisfied: numpy>=1.22.4 in c:\users\mohan\anaconda3\lib\site-packages (from pandas) (1.24.3)
Requirement already satisfied: six>=1.5 in c:\users\mohan\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
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WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
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WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
```

In [5]:

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\mohan\anaconda3\lib\site-packages (3.4.3)
Requirement already satisfied: numpy>=1.16 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (1.24.3)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (2.8.2)
Requirement already satisfied: pillow>=6.2.0 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (9.5.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (1.3.1)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
Requirement already satisfied: pyparsing>=2.2.1 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (3.0.4)
Requirement already satisfied: cycler>=0.10 in c:\users\mohan\anaconda3\lib\site-packages (from matplotlib) (0.10.0)
Requirement already satisfied: six in c:\users\mohan\anaconda3\lib\site-packages (from cycler>=0.10->matplotlib) (1.16.0)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -atplotlib (c:\users\mohan\anaconda3\lib\site-packages)
```

In [22]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sb
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
```

```

from sklearn.linear_model import LinearRegression
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.cluster import KMeans
from sklearn.metrics import confusion_matrix
from sklearn.metrics import mean_squared_error

```

```

%matplotlib inline
import warnings
warnings.filterwarnings("ignore")

```

In [56]: `data=pd.read_csv("student_data.csv")`

In [57]: `data.head()`

Out[57]:

|   | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Fees   |
|---|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|--------|
| 0 | 165527     | Bryan Rogers    | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 155152 |
| 1 | 635763     | James Hogan     | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 157870 |
| 2 | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 55662  |
| 3 | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 134955 |
| 4 | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 125934 |

In [58]: `data.tail()`

Out[58]:

|        | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Fees |
|--------|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|------|
| 199995 | 220556     | Kelly Bradley   | 2001-06-06    | Electrical Engineering | 2017              | 2021                        | 4                | Data Science     | 1641 |
| 199996 | 957592     | Jacob Smith     | 2003-12-15    | Electrical Engineering | 2017              | 2019                        | 3                | Web Development  | 1351 |
| 199997 | 786798     | Nicole Davis    | 2000-07-27    | Chemical Engineering   | 2022              | 2023                        | 1                | Data Science     | 1394 |
| 199998 | 142498     | Natasha Rosales | 2001-02-16    | Civil Engineering      | 2017              | 2018                        | 1                | Data Science     | 731  |
| 199999 | 930898     | Charles Marsh   | 2005-10-17    | Computer Science       | 2018              | 2020                        | 2                | Machine Learning | 755  |

In [59]: `data.shape`

Out[59]: (200000, 10)

In [60]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Student ID                            200000 non-null int64
1   Student Name                          200000 non-null object
2   Date of Birth                         200000 non-null object
3   Field of Study                        200000 non-null object
4   Year of Admission                     200000 non-null int64
5   Expected Year of Graduation           200000 non-null int64
6   Current Semester                     200000 non-null int64
7   Specialization                       200000 non-null object
8   Fees                                 200000 non-null int64
9   Discount on Fees                     200000 non-null int64
dtypes: int64(6), object(4)
memory usage: 15.3+ MB
```

In [61]: `data.describe()`

Out[61]:

|              | Student ID    | Year of Admission | Expected Year of Graduation | Current Semester | Fees          | Discount on Fees |
|--------------|---------------|-------------------|-----------------------------|------------------|---------------|------------------|
| <b>count</b> | 200000.000000 | 200000.000000     | 200000.000000               | 200000.000000    | 200000.000000 | 200000.000000    |
| <b>mean</b>  | 549367.492925 | 2018.997685       | 2019.995235                 | 2.499020         | 125092.847595 | 12484.258575     |
| <b>std</b>   | 259361.565011 | 2.002381          | 1.997744                    | 1.117804         | 43287.894903  | 8788.362629      |
| <b>min</b>   | 100001.000000 | 2016.000000       | 2017.000000                 | 1.000000         | 50000.000000  | 0.000000         |
| <b>25%</b>   | 325311.000000 | 2017.000000       | 2018.000000                 | 1.000000         | 87641.500000  | 5383.000000      |
| <b>50%</b>   | 548855.500000 | 2019.000000       | 2020.000000                 | 2.000000         | 125221.000000 | 10792.500000     |
| <b>75%</b>   | 774182.500000 | 2021.000000       | 2022.000000                 | 3.000000         | 162597.250000 | 18154.000000     |
| <b>max</b>   | 999997.000000 | 2022.000000       | 2023.000000                 | 4.000000         | 200000.000000 | 39865.000000     |

In [62]: `# to cheak the null value`  
`data.isnull().sum()`

Out[62]:

|                             |       |
|-----------------------------|-------|
| Student ID                  | 0     |
| Student Name                | 0     |
| Date of Birth               | 0     |
| Field of Study              | 0     |
| Year of Admission           | 0     |
| Expected Year of Graduation | 0     |
| Current Semester            | 0     |
| Specialization              | 0     |
| Fees                        | 0     |
| Discount on Fees            | 0     |
| dtype:                      | int64 |

In [63]: `# to count the value of any spacific columns`  
`data['Year of Admission'].value_counts()`



```
Out[63]: Year of Admission
2017    28760
2022    28703
2016    28646
2019    28618
2021    28483
2018    28435
2020    28355
Name: count, dtype: int64
```

```
In [64]: # to change the columns name

data.rename(columns={'Fees': 'Annual_Fess'}, inplace=True)
```

```
In [65]: data
```

```
Out[65]:
```

|        | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Annual Fees |
|--------|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-------------|
| 0      | 165527     | Bryan Rogers    | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 28760       |
| 1      | 635763     | James Hogan     | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 28703       |
| 2      | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 28646       |
| 3      | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 28618       |
| 4      | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 28483       |
| ...    | ...        | ...             | ...           | ...                    | ...               | ...                         | ...              | ...              | ...         |
| 199995 | 220556     | Kelly Bradley   | 2001-06-06    | Electrical Engineering | 2017              | 2021                        | 4                | Data Science     | 28435       |
| 199996 | 957592     | Jacob Smith     | 2003-12-15    | Electrical Engineering | 2017              | 2019                        | 3                | Web Development  | 28355       |
| 199997 | 786798     | Nicole Davis    | 2000-07-27    | Chemical Engineering   | 2022              | 2023                        | 1                | Data Science     |             |
| 199998 | 142498     | Natasha Rosales | 2001-02-16    | Civil Engineering      | 2017              | 2018                        | 1                | Data Science     |             |
| 199999 | 930898     | Charles Marsh   | 2005-10-17    | Computer Science       | 2018              | 2020                        | 2                | Machine Learning |             |

200000 rows × 10 columns



```
In [68]: data.head(2)
```

Out[68]:

|   | Student ID | Student Name | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Annual_Fe |
|---|------------|--------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-----------|
| 0 | 165527     | Bryan Rogers | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 1551      |
| 1 | 635763     | James Hogan  | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 1578      |



In [69]:

```
# to cheak the duplicate value
duplicate=data.duplicated()
```

In [70]:

```
duplicate
```

Out[70]:

```
0      False
1      False
2      False
3      False
4      False
...
199995  False
199996  False
199997  False
199998  False
199999  False
Length: 200000, dtype: bool
```

In [71]:

```
# create a new column
data['Present']='present'
```

In [72]:

```
data.head()
```

Out[72]:

|   | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Annual_F |
|---|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|----------|
| 0 | 165527     | Bryan Rogers    | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 1551     |
| 1 | 635763     | James Hogan     | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 1578     |
| 2 | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 556      |
| 3 | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 1349     |
| 4 | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 1259     |



In [73]:

```
# change type object to to_datetime
```

```
data['Date of Birth']=pd.to_datetime(data['Date of Birth'])
```

In [74]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Student ID                           200000 non-null  int64
1   Student Name                         200000 non-null  object
2   Date of Birth                        200000 non-null  datetime64[ns]
3   Field of Study                       200000 non-null  object
4   Year of Admission                    200000 non-null  int64
5   Expected Year of Graduation          200000 non-null  int64
6   Current Semester                    200000 non-null  int64
7   Specialization                      200000 non-null  object
8   Annual_Fess                         200000 non-null  int64
9   Discount on Fees                    200000 non-null  int64
10  Present                             200000 non-null  object
dtypes: datetime64[ns](1), int64(6), object(4)
memory usage: 16.8+ MB
```

In [75]:

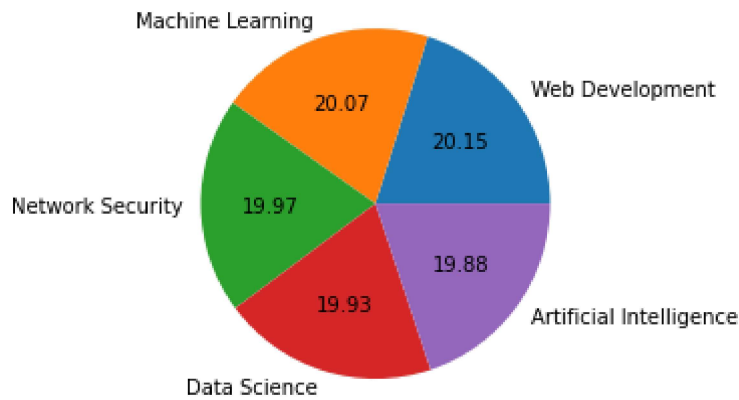
```
data.head()
```

Out[75]:

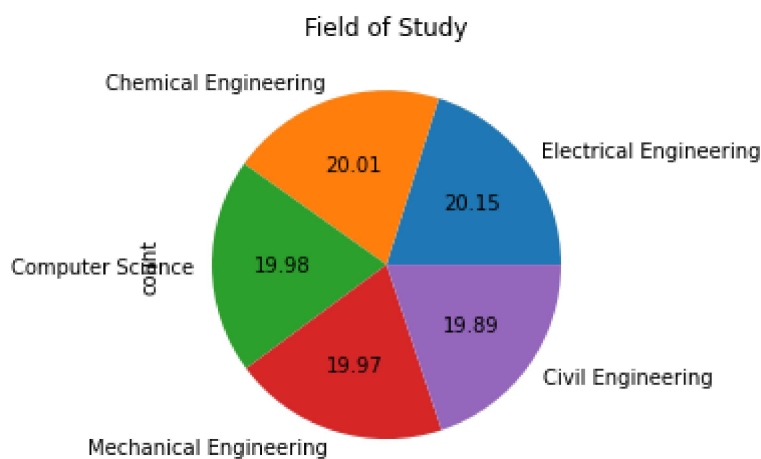
|   | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | Annual_F |
|---|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|----------|
| 0 | 165527     | Bryan Rogers    | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 155      |
| 1 | 635763     | James Hogan     | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 1578     |
| 2 | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 556      |
| 3 | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 1349     |
| 4 | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 1259     |

In [76]:

```
weightage = data['Specialization'].value_counts()
py.pie(weightage, labels=['Web Development', 'Machine Learning', 'Network Security', 'Data Science'])
py.show()
```



```
In [77]: data['Field of Study'].value_counts().plot(kind='pie', autopct='%.2f')
py.title('Field of Study')
py.show()
```



```
In [78]: data['Field of Study'].value_counts()
```

```
Out[78]: Field of Study
Electrical Engineering    40305
Chemical Engineering      40020
Computer Science          39960
Mechanical Engineering     39941
Civil Engineering         39774
Name: count, dtype: int64
```

```
In [79]: clean_df = data[data['Year of Admission'] < data['Expected Year of Graduation']]
```

```
In [80]: pd.crosstab(clean_df['Year of Admission'], clean_df['Field of Study'])
```

```
Out[80]:
```

| Field of Study    | Chemical Engineering | Civil Engineering | Computer Science | Electrical Engineering | Mechanical Engineering |
|-------------------|----------------------|-------------------|------------------|------------------------|------------------------|
| Year of Admission |                      |                   |                  |                        |                        |
| 2016              | 5728                 | 5755              | 5763             | 5745                   | 5655                   |
| 2017              | 4967                 | 4970              | 4855             | 4986                   | 4923                   |
| 2018              | 4076                 | 4034              | 4061             | 4069                   | 4124                   |



| Field of Study    | Chemical Engineering | Civil Engineering | Computer Science | Electrical Engineering | Mechanical Engineering |
|-------------------|----------------------|-------------------|------------------|------------------------|------------------------|
| Year of Admission |                      |                   |                  |                        |                        |
| 2019              | 3310                 | 3204              | 3163             | 3304                   | 3333                   |
| 2020              | 2458                 | 2412              | 2330             | 2430                   | 2454                   |
| 2021              | 1653                 | 1605              | 1620             | 1653                   | 1644                   |
| 2022              | 817                  | 809               | 826              | 796                    | 821                    |

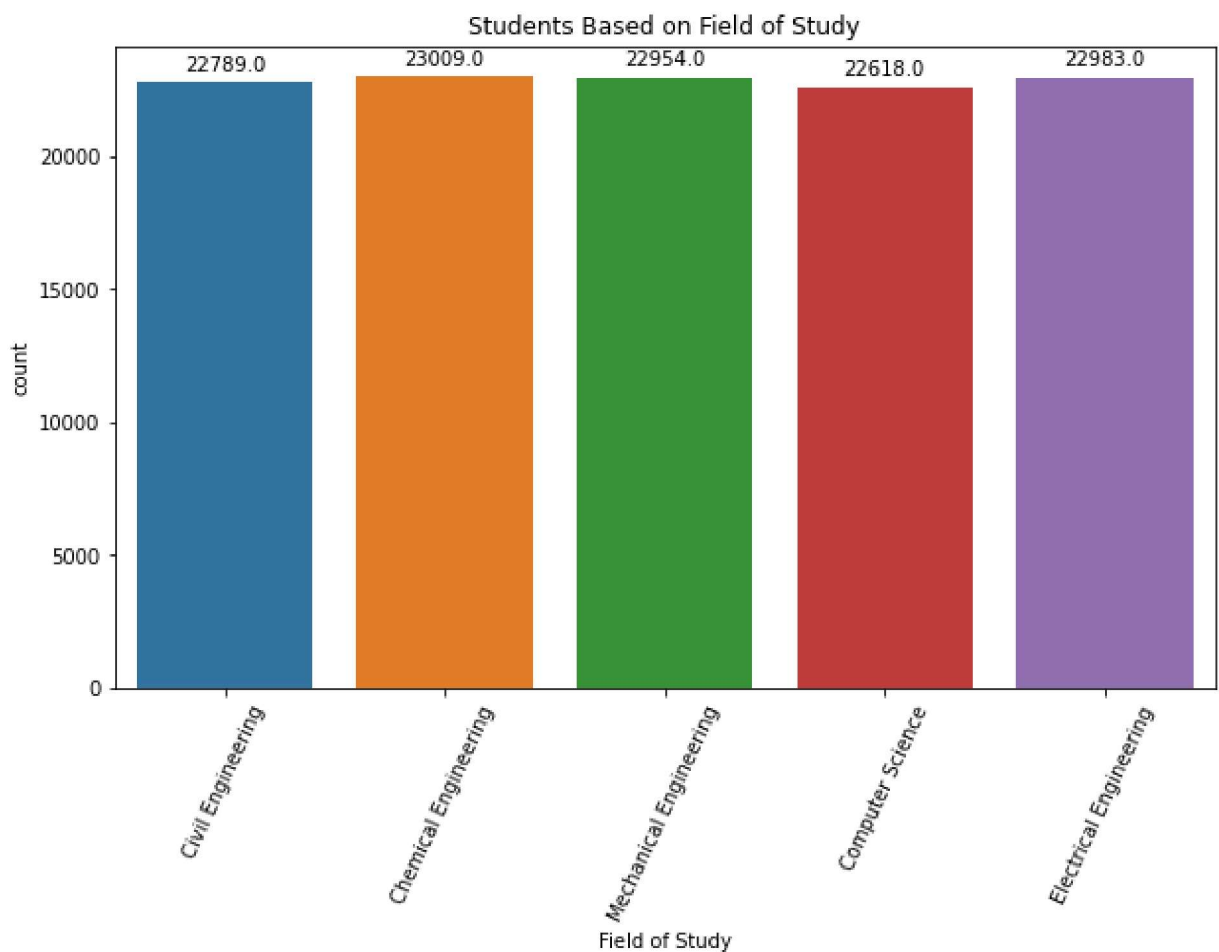
In [81]:

```

py.figure(figsize=(10,6))
py.xticks(rotation=65)
plot = sb.countplot(x=clean_df['Field of Study'])
for p in plot.patches:
    plot.annotate(format(p.get_height(), '.1f'),
                  (p.get_x() + p.get_width() / 2., p.get_height()),
                  ha = 'center', va = 'center',
                  xytext = (0, 9),
                  textcoords = 'offset points')
py.title('Students Based on Field of Study')

```

Out[81]: Text(0.5, 1.0, 'Students Based on Field of Study')



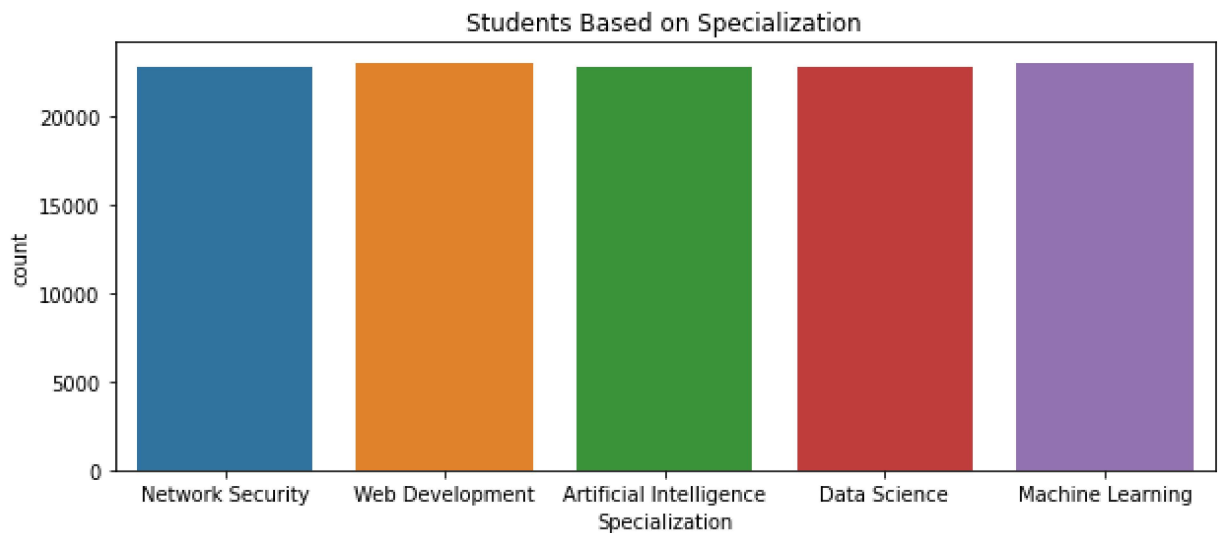
In [82]:

```

py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Specialization'])
py.title('Students Based on Specialization')

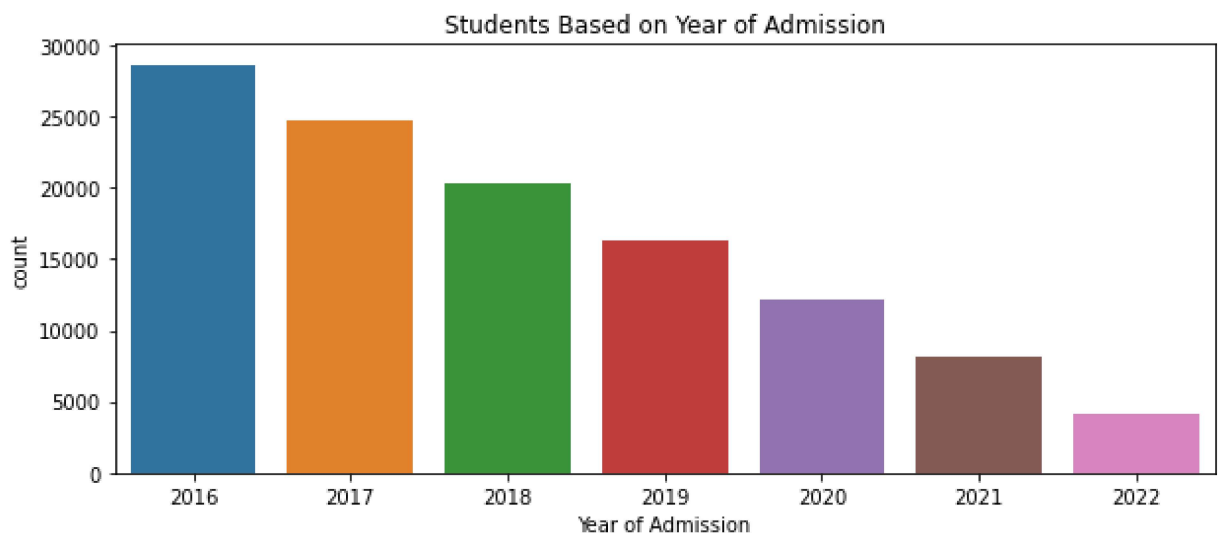
```

Out[82]: Text(0.5, 1.0, 'Students Based on Specialization')



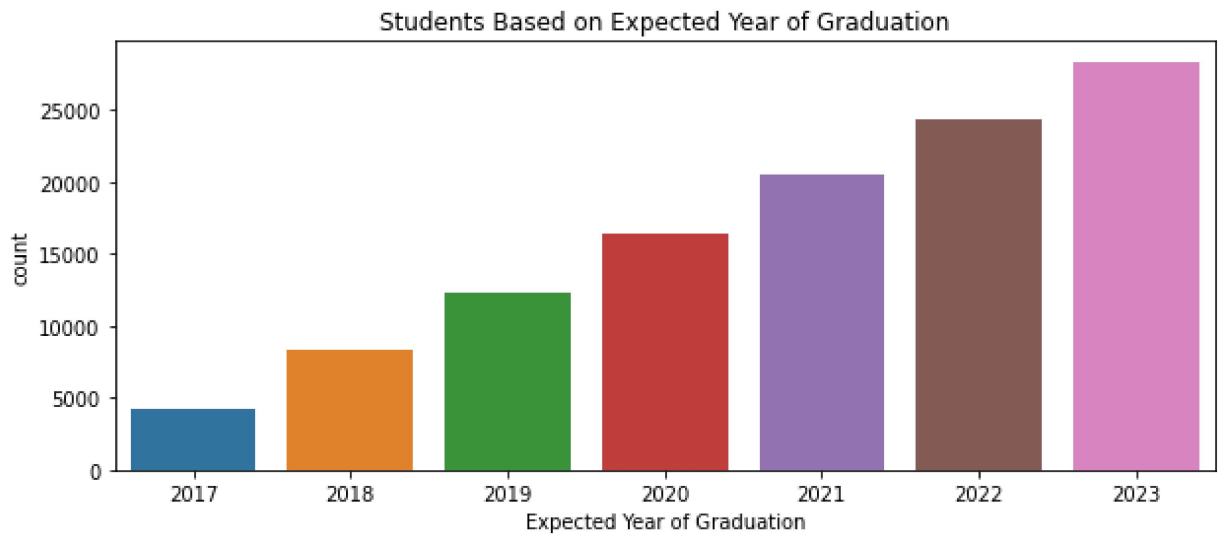
```
In [83]: py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Year of Admission'])
py.title('Students Based on Year of Admission')
```

Out[83]: Text(0.5, 1.0, 'Students Based on Year of Admission')



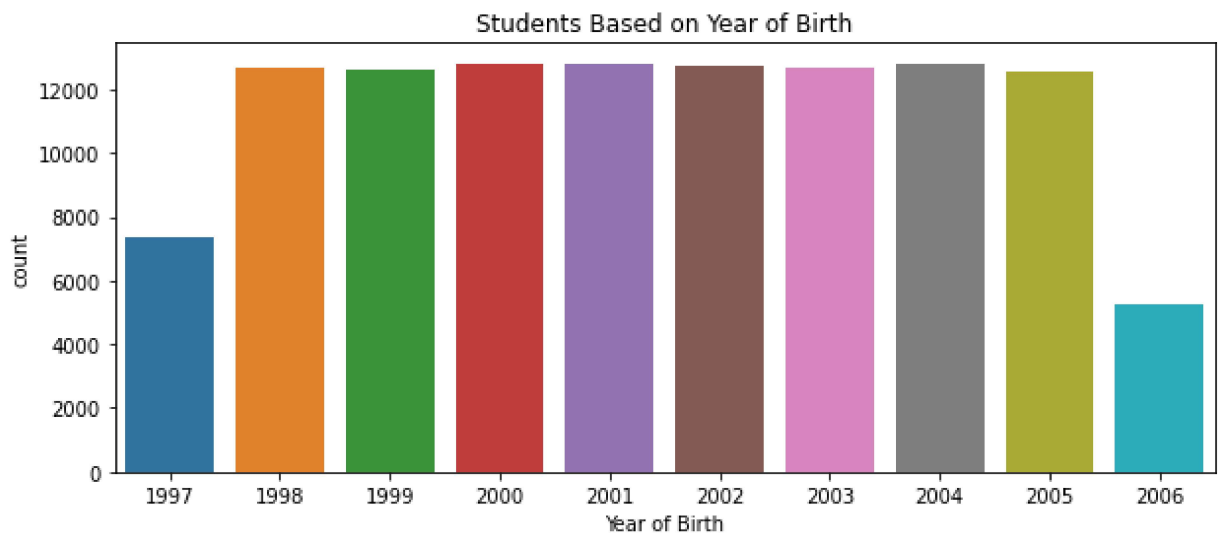
```
In [84]: py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Expected Year of Graduation'])
py.title('Students Based on Expected Year of Graduation')
```

Out[84]: Text(0.5, 1.0, 'Students Based on Expected Year of Graduation')



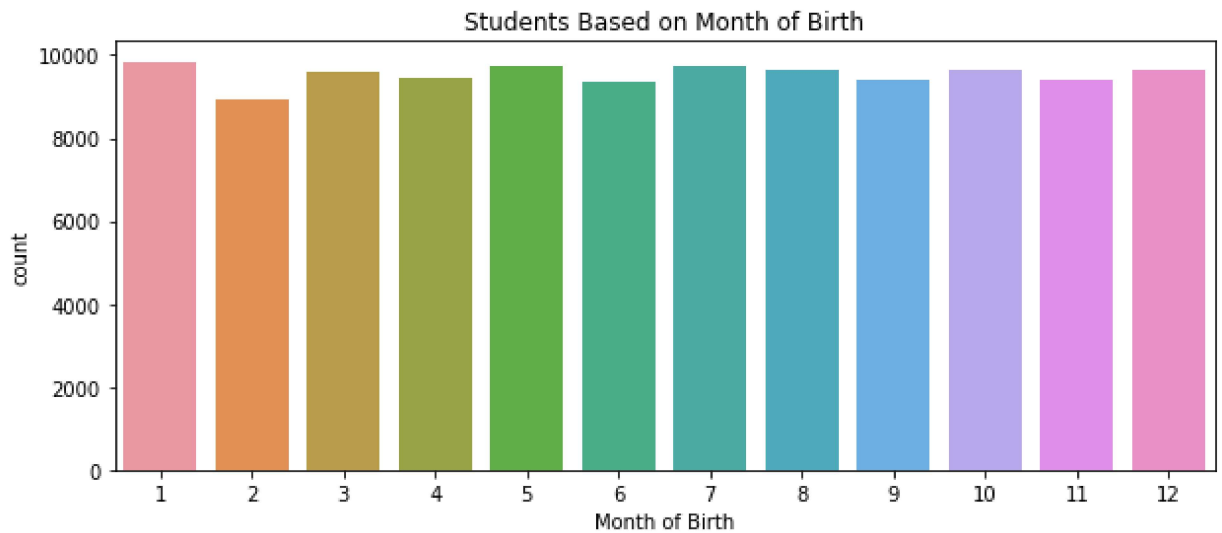
```
In [85]: py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Date of Birth'].dt.year)
py.title('Students Based on Year of Birth')
py.xlabel('Year of Birth')
```

Out[85]: Text(0.5, 0, 'Year of Birth')



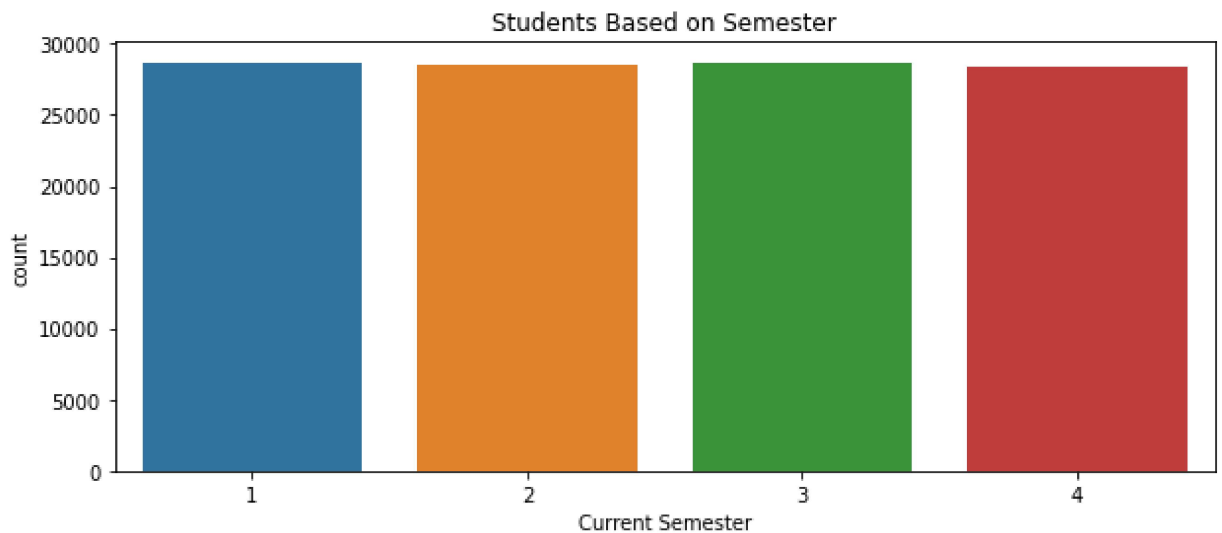
```
In [86]: py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Date of Birth'].dt.month)
py.title('Students Based on Month of Birth')
py.xlabel('Month of Birth')
```

Out[86]: Text(0.5, 0, 'Month of Birth')



```
In [87]: py.figure(figsize=(10,4))
sb.countplot(x=clean_df['Current Semester'])
py.title('Students Based on Semester')
```

```
Out[87]: Text(0.5, 1.0, 'Students Based on Semester')
```

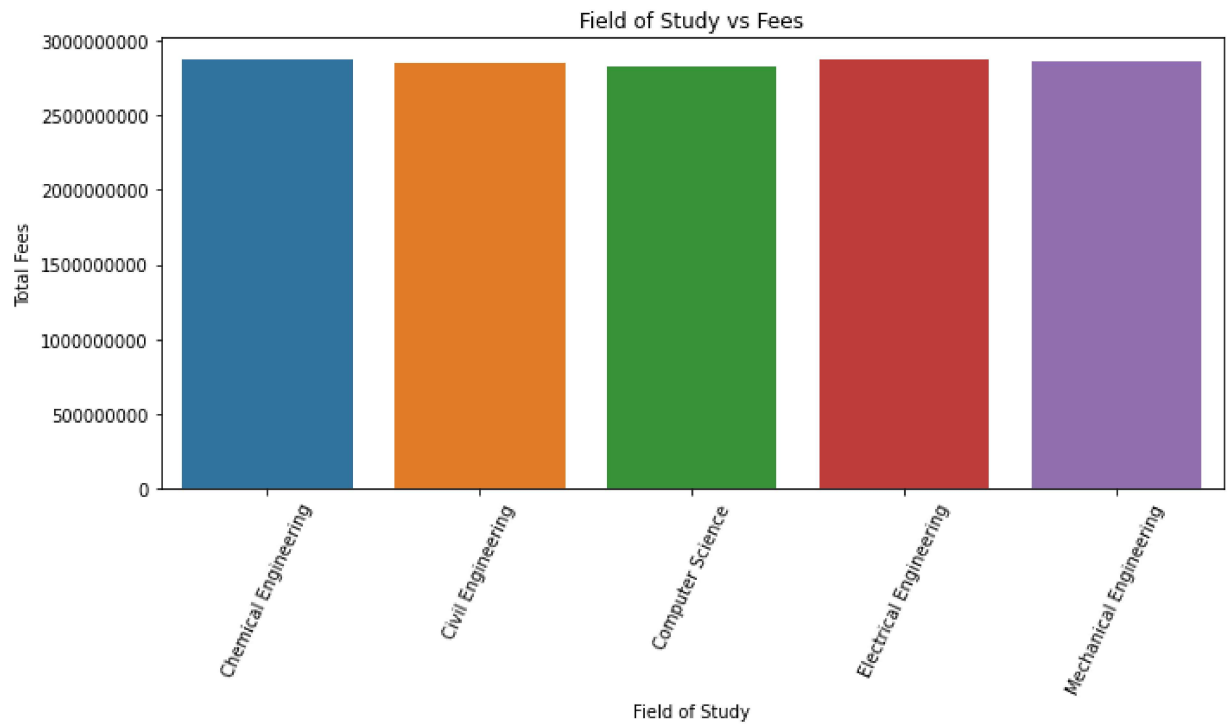


```
In [88]: py.figure(figsize=(10, 6))
py.ticklabel_format(style='plain', axis='y')

# Group and reset index for seaborn
fees_by_field = clean_df.groupby('Field of Study')['Annual_Fess'].sum().reset_index()

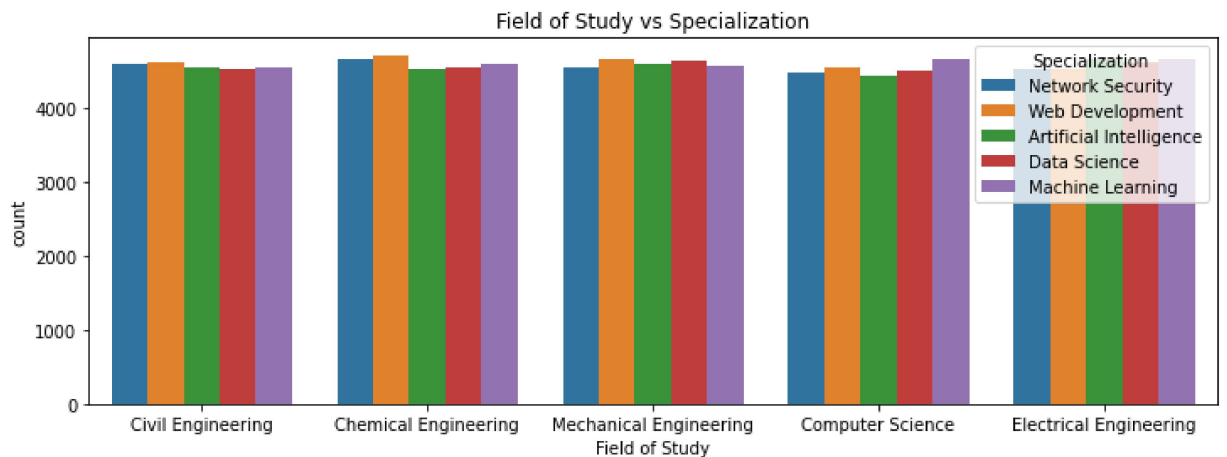
# Barplot
sb.barplot(x='Field of Study', y='Annual_Fess', data=fees_by_field)

py.xticks(rotation=65)
py.title('Field of Study vs Fees')
py.xlabel('Field of Study')
py.ylabel('Total Fees')
py.tight_layout()
py.show()
```



```
In [89]: py.figure(figsize=(12,4))
sb.countplot(x=clean_df['Field of Study'], hue=clean_df['Specialization'])
py.title('Field of Study vs Specialization')
```

Out[89]: Text(0.5, 1.0, 'Field of Study vs Specialization')



```
In [90]: # to change the columns name

data.rename(columns={'Annual_Fess':'Fees'},inplace=True)
```

In [91]: data

Out[91]:

|   | Student ID | Student Name | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | I   |
|---|------------|--------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-----|
| 0 | 165527     | Bryan Rogers | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 155 |
| 1 | 635763     | James Hogan  | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 157 |

|        | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | I   |
|--------|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-----|
| 2      | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 55  |
| 3      | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 134 |
| 4      | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 125 |
| ...    | ...        | ...             | ...           | ...                    | ...               | ...                         | ...              | ...              | ... |
| 199995 | 220556     | Kelly Bradley   | 2001-06-06    | Electrical Engineering | 2017              | 2021                        | 4                | Data Science     | 164 |
| 199996 | 957592     | Jacob Smith     | 2003-12-15    | Electrical Engineering | 2017              | 2019                        | 3                | Web Development  | 135 |
| 199997 | 786798     | Nicole Davis    | 2000-07-27    | Chemical Engineering   | 2022              | 2023                        | 1                | Data Science     | 139 |
| 199998 | 142498     | Natasha Rosales | 2001-02-16    | Civil Engineering      | 2017              | 2018                        | 1                | Data Science     | 73  |
| 199999 | 930898     | Charles Marsh   | 2005-10-17    | Computer Science       | 2018              | 2020                        | 2                | Machine Learning | 75  |

200000 rows × 11 columns

In [92]:

```
data["day"] = data["Date of Birth"].dt.day
data["month"] = data["Date of Birth"].dt.month
data["year"] = data["Date of Birth"].dt.year
```

In [93]:

```
data
```

Out[93]:

|        | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | I   |
|--------|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-----|
| 0      | 165527     | Bryan Rogers    | 2006-01-19    | Computer Science       | 2020              | 2017                        | 3                | Web Development  | 155 |
| 1      | 635763     | James Hogan     | 1999-05-23    | Mechanical Engineering | 2020              | 2020                        | 2                | Machine Learning | 157 |
| 2      | 740021     | David Robinson  | 1997-12-02    | Civil Engineering      | 2017              | 2022                        | 1                | Network Security | 55  |
| 3      | 433076     | Susan Miller    | 1999-10-30    | Computer Science       | 2021              | 2019                        | 1                | Data Science     | 134 |
| 4      | 441628     | Brittany Martin | 1998-01-10    | Chemical Engineering   | 2016              | 2018                        | 1                | Network Security | 125 |
| ...    | ...        | ...             | ...           | ...                    | ...               | ...                         | ...              | ...              | ... |
| 199995 | 220556     | Kelly Bradley   | 2001-06-06    | Electrical Engineering | 2017              | 2021                        | 4                | Data Science     | 164 |



|        | Student ID | Student Name    | Date of Birth | Field of Study         | Year of Admission | Expected Year of Graduation | Current Semester | Specialization   | I   |
|--------|------------|-----------------|---------------|------------------------|-------------------|-----------------------------|------------------|------------------|-----|
| 199996 | 957592     | Jacob Smith     | 2003-12-15    | Electrical Engineering | 2017              | 2019                        | 3                | Web Development  | 135 |
| 199997 | 786798     | Nicole Davis    | 2000-07-27    | Chemical Engineering   | 2022              | 2023                        | 1                | Data Science     | 139 |
| 199998 | 142498     | Natasha Rosales | 2001-02-16    | Civil Engineering      | 2017              | 2018                        | 1                | Data Science     | 73  |
| 199999 | 930898     | Charles Marsh   | 2005-10-17    | Computer Science       | 2018              | 2020                        | 2                | Machine Learning | 75  |

200000 rows × 14 columns

```
In [94]: print("Unique Values in field col")
print(data["Field of Study"].unique(), end="\n\n")    # returns unique values in fie

print("Unique Values in specialization col")
print(data["Specialization"].unique())                # returns unique values in specializati
```

Unique Values in field col

```
['Computer Science' 'Mechanical Engineering' 'Civil Engineering'
 'Chemical Engineering' 'Electrical Engineering']
```

Unique Values in specialization col

```
['Web Development' 'Machine Learning' 'Network Security' 'Data Science'
 'Artificial Intelligence']
```

```
In [95]: cross_tab = pd.crosstab(data["Field of Study"], data["Specialization"])
cross_tab
```

Out[95]:

| Specialization         | Artificial Intelligence | Data Science | Machine Learning | Network Security | Web Development |
|------------------------|-------------------------|--------------|------------------|------------------|-----------------|
| Field of Study         |                         |              |                  |                  |                 |
| Chemical Engineering   | 7945                    | 7955         | 7924             | 8040             | 8156            |
| Civil Engineering      | 7864                    | 7925         | 7880             | 8076             | 8029            |
| Computer Science       | 7900                    | 8018         | 8131             | 7887             | 8024            |
| Electrical Engineering | 8058                    | 8032         | 8201             | 8028             | 7986            |
| Mechanical Engineering | 7997                    | 7940         | 8006             | 7901             | 8097            |

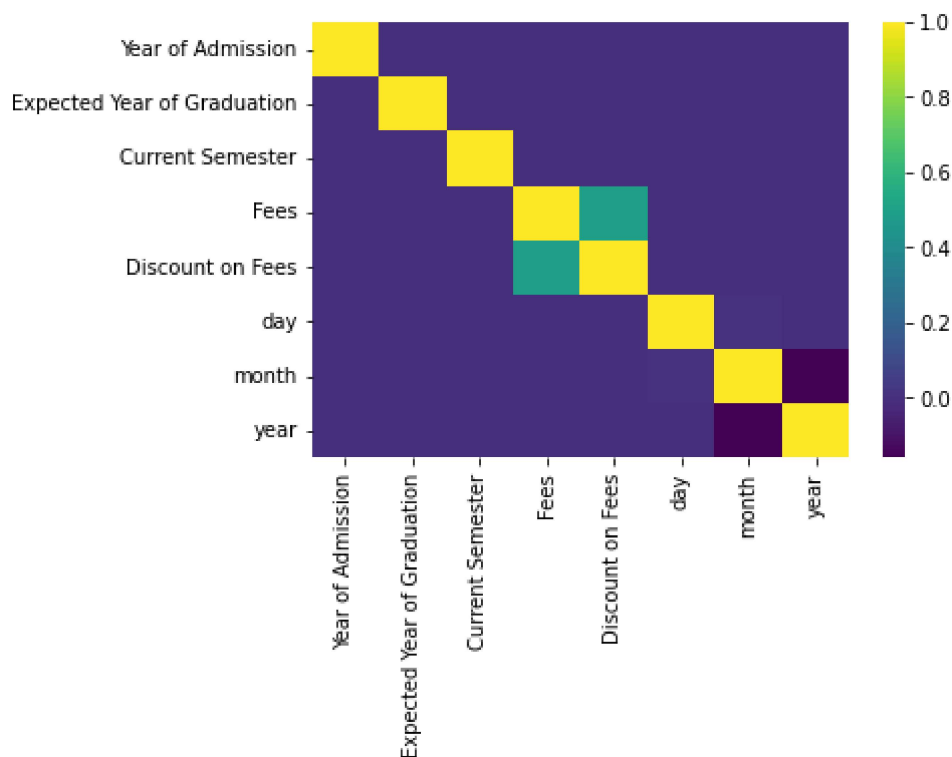
```
In [96]: df = data.drop("Student ID", axis = 1)
```

```
In [97]: if df.isnull().values.any():
print("yess")
else:
print("no")
```

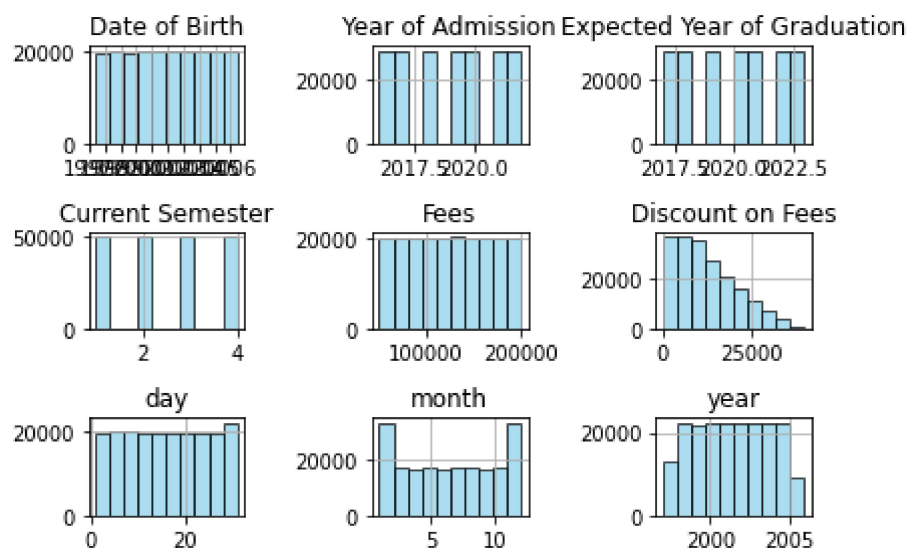
no

```
In [98]: corr = df.corr(numeric_only=True)
sns.heatmap(corr, cmap="viridis")
```

```
Out[98]: <AxesSubplot:>
```



```
In [99]: df.hist(bins=10, color='skyblue', edgecolor='black', alpha=0.7)
py.tight_layout()
```



```
In [ ]:
```